Structural Break and Time Series Modeling of Weekly Cereal Sales

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Introduction

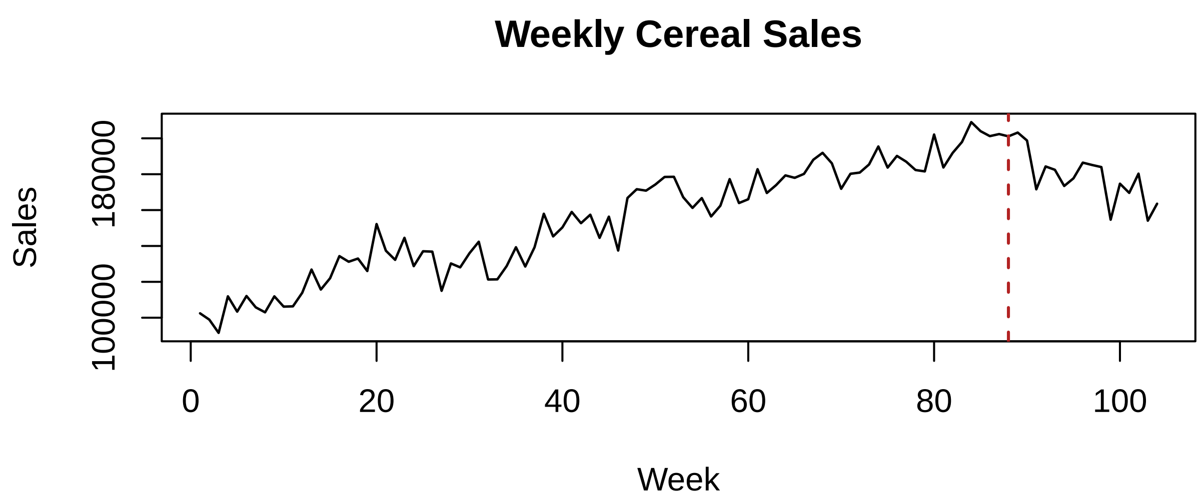
This report examines a published example containing 104 weeks of cereal sales. In the scenario accompanying the data, a competing product enters the market in week 88. The analysis asks whether the sales level or weekly trend changed at that point while accounting for serial correlation.

The dataset comes from the weekly cereal-sales intervention example in Montgomery, Jennings, and Kulahci's *Introduction to Time Series Analysis and Forecasting*, Second Edition, and is available through Wiley's companion materials. The source does not identify the brands or establish whether the values are observed or simulated.

The analysis has three goals:

1. Estimate the immediate level change and the change in trend at week 88.
2. Diagnose and model autocorrelation in the regression residuals.
3. Produce short-term model-based forecasts and describe their limitations.

Data and Exploratory Analysis



Sales rise through most of the series and then decline after week 88. A segmented trend is therefore a natural description of the visible change. The event timing is supplied with the example, so the analysis treats week 88 as known rather than searching the series for a break point.

The design is an interrupted time series without a control series or additional covariates. It estimates a change in the brand's weekly sales associated with the stated event date, but it cannot establish that competitor entry caused the change; market share is not observed.

Segmented Regression

Let I_t equal one beginning in week 88, and let $A_t = \max(0, t - 88)$. The deterministic part of the model is

$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 A_t + e_t.$$

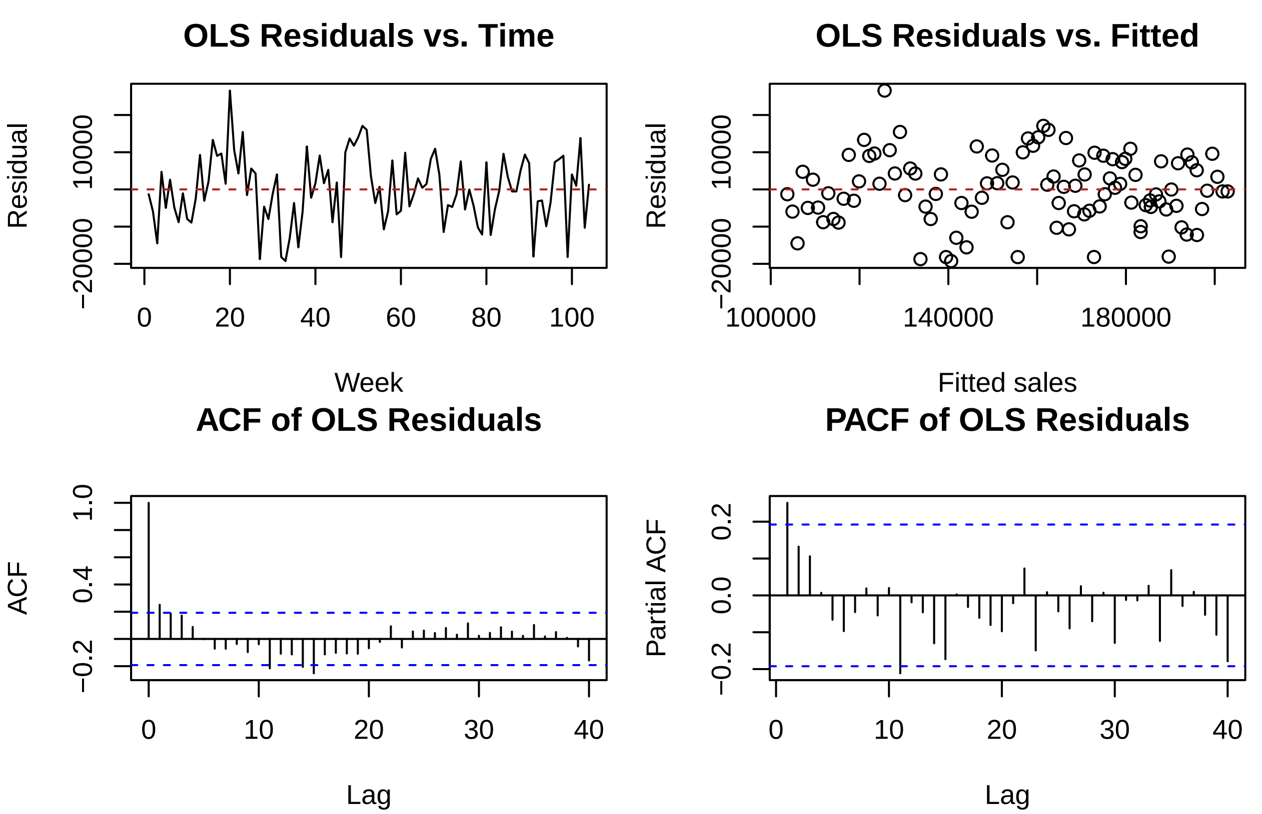
This centered parameterization makes the quantities of interest explicit:

- β_1 is the pre-entry weekly trend.
- β_2 is the immediate level shift at week 88 relative to the pre-entry trend.
- β_3 is the change in weekly trend.
- $\beta_1 + \beta_3$ is the post-entry weekly trend.

The analysis uses a two-step workflow. The segmented regression estimates the level and trend changes, and an ARMA(1,1) model captures the remaining serial dependence in the regression residuals.

Regression quantity	Estimate
Pre-entry weekly trend	1,153
Immediate level shift at week 88	-8,118
Change in weekly trend	-3,257
Post-entry weekly trend	-2,104

Diagnostic	p-value
Ramsey RESET	0.521
Durbin-Watson (positive autocorrelation)	0.002
Nonconstant variance score test	0.291
Shapiro-Wilk	0.514
KPSS level stationarity	>0.100



The fitted line rises by approximately 1,153 units per week before entry and declines by approximately 2,104 units per week afterward, a change in slope of -3,257 units per week. The immediate level-shift estimate is -8,118 units. The primary modeled change is the reversal in trend rather than a one-time shift.

The Durbin-Watson result and the lag-one autocorrelation show that independence is not a reasonable OLS assumption. The other tests and plots do not flag strong nonlinearity, nonconstant variance, or nonnormality, but those p-values are exploratory because they were computed from dependent residuals. The KPSS test found no evidence that the regression residuals are nonstationary ($p > 0.1$), so they are modeled without differencing.

Only two annual cycles are observed. The displayed ACF and PACF cover lags through 40, so annual seasonality at lag 52 cannot be assessed reliably from this sample. No seasonal terms are included.

Residual Time Series Modeling

The second step models the autocorrelation left in the OLS residuals. AR(1), MA(1), and ARMA(1,1) models are compared because the residual ACF and PACF concentrate most of their structure at short lags. The residual mean is fixed at zero because the regression includes an intercept.

Residual model	AIC	BIC	Ljung-Box p-value (lag 10)
AR(1)	2188.0	2193.3	0.836
MA(1)	2189.4	2194.7	0.620
ARMA(1,1)	2187.6	2195.6	0.955

ARMA(1,1) has the smallest AIC, while BIC favors the simpler AR(1). Both leave no evident short-lag autocorrelation. ARMA(1,1) is retained for the residual stage because the analysis includes forecasting, both fitted terms contribute within that residual model, and its AIC is lowest. The AIC difference from AR(1) is less than one point, so AR(1) remains a reasonable parsimonious alternative rather than a decisively inferior model.

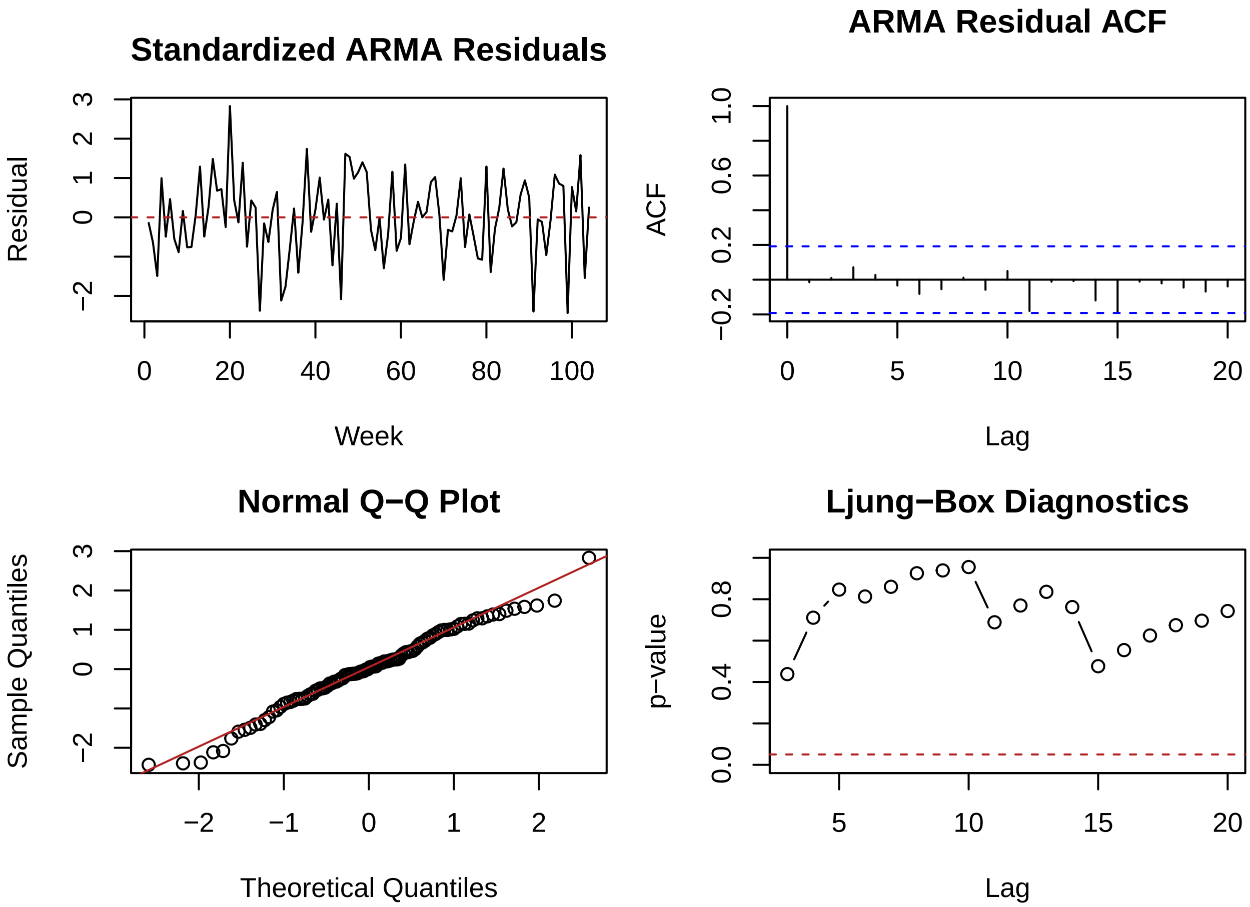
Residual parameter	Estimate	Standard error	p-value
AR(1)	0.673	0.175	<0.001
MA(1)	-0.451	0.203	0.027

The two fitted stages are

$$
\begin{aligned}
\widehat{y}_t &= 102{,}585 + 1{,}153t - 8{,}118I_t - 3{,}257A_t,\\
r_t &= y_t - \widehat{y}_t,\\
r_t &= 0.673r_{t-1} - 0.451\varepsilon_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2).
\end{aligned}
$$

The first equation gives the fitted regression value at week t, and r_t is the corresponding residual. The ARMA(1,1) model captures the serial dependence in those residuals. Diagnostics of the resulting ARMA residuals show no evidence of remaining autocorrelation, strong nonnormality, or conditional heteroskedasticity.

Residual-model diagnostic	p-value
Ljung-Box ARMA residual autocorrelation (lag 10)	0.955
Shapiro-Wilk normality	0.385
ARCH LM conditional heteroskedasticity (5 lags)	0.823

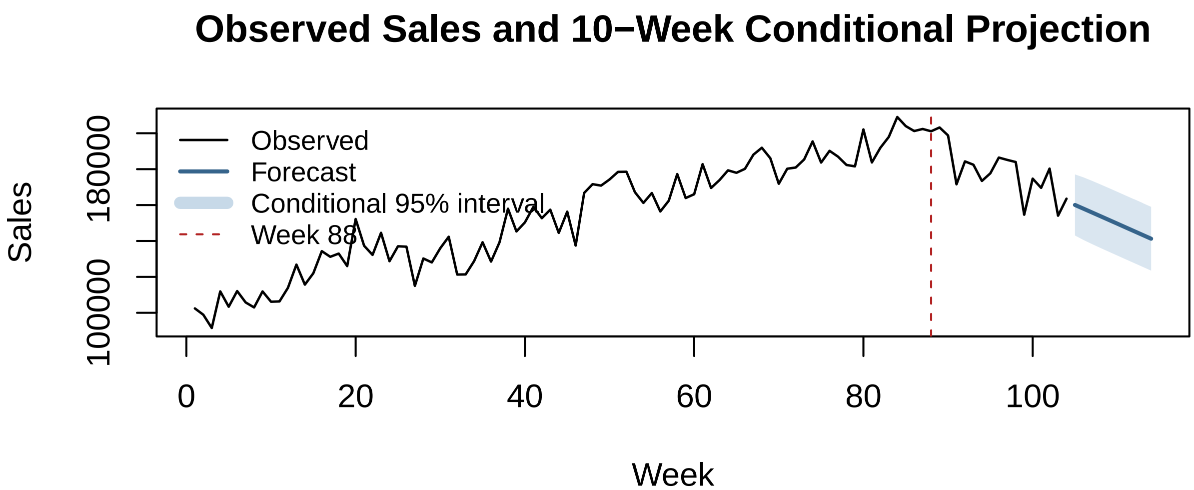


The lag-10 Ljung-Box test finds no evidence of remaining autocorrelation through lag 10 ($p = 0.955$). The Shapiro-Wilk test and Q-Q plot do not flag strong nonnormality ($p = 0.385$), and the five-lag ARCH test does not flag conditional heteroskedasticity ($p = 0.823$). The fitted AR and MA roots satisfy the stationarity and invertibility conditions.

Forecasting

The point forecast adds the segmented-regression projection to the forecast of its ARMA(1,1) residual. These projections assume that the post-entry trend and residual process continue unchanged. The intervals reflect uncertainty from the residual process only; they treat the regression coefficients and fitted ARMA parameters as fixed and exclude model-selection uncertainty.

Week	Forecast	Cond. 95% lower	Cond. 95% upper
105	160,065	143,045	177,085
106	158,000	140,568	175,433
107	155,923	138,307	173,539
108	153,837	136,139	171,535
109	151,745	134,010	169,481
110	149,650	131,897	167,402
111	147,551	129,791	165,311
112	145,451	127,688	163,214
113	143,350	125,585	161,115
114	141,248	123,483	159,013



As a limited forecast check, both stages were refit at ten rolling origins to predict weeks 95 through 104 one step ahead. The segmented regression is identical across the two candidates; only the fitted residual model changes.

Measure	Result
Two-step AR(1) RMSE	13,004
Two-step ARMA(1,1) RMSE	12,903
Naive RMSE	15,107
AR(1) residual-only interval coverage	8 of 10 forecasts
ARMA(1,1) residual-only interval coverage	8 of 10 forecasts

Both two-step forecasts outperform the naive benchmark in this short window. ARMA(1,1) is 101 RMSE units (0.8%) below AR(1), which is too small a difference to distinguish the residual specifications decisively. Each covers 8 of 10 residual-only intervals. Because the evaluation contains only 10 forecasts—with 87 pre-entry and 7 post-entry observations at the first origin—and the intervals omit regression and parameter-estimation uncertainty, the projection is illustrative rather than a validated long-range forecast.

Conclusion

The segmented regression estimates a pre-entry rise of 1,153 units per week and a post-entry decline of 2,104 units per week, with a slope change of -3,257 units per week. The estimated immediate level shift is -8,118 units; the primary modeled change is the trend reversal rather than a one-time jump.

An ARMA(1,1) model removes the evident short-lag autocorrelation from the regression residuals. AR(1) is also adequate, and the limited rolling check does not establish a meaningful forecasting advantage for either residual model. This uncontrolled example with a short post-entry period supports a descriptive association with the stated event date, not a causal conclusion; market share is not observed.

Data Source

Montgomery, D. C., Jennings, C. L., and Kulahci, M. (2015). *Introduction to Time Series Analysis and Forecasting* (2nd ed.). Wiley. Weekly cereal-sales intervention example and companion data.